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Report: The Jeep Cherokee Hack Case (2015)

1. Introduction

The development of new technologies has changed how modern vehicles operate. Today's vehicles have changed from being all-mechanical to now having software, sensors, and Internet connectivity to be able to operate with more convenience, safety, and performance.

Vehicle features, including GPS navigation, remote diagnostics, infotainment systems, and wireless updates, have afforded vehicles the ability to operate intelligently and in a user-friendly manner. However, the increase in vehicle connectivity is also related to increased cybersecurity risk. As vehicles become part of the Internet of Things, they are subjected to many vulnerabilities that attackers can use to gain unauthorized access to vehicles.

The Jeep Cherokee hack from 2015 represents an event where vehicle digital systems can be compromised remotely and can create a potential for significant harm to passenger safety and to public safety. The Jeep incident was pivotal for the automotive industry, which made the urgent need for the automotive industry to address cybersecurity as a key element of vehicle design and operation.

This report will cover the Jeep Cherokee hack from a social, ethical, legal, and professional perspective. It will also provide a personal reflection and an ethical decision-making framework for the engineering and computing professions in relation to the incident discussed.

2. Background of the Scenario

In July 2015 a demonstration conducted by cybersecurity researchers Charlie Miller and Chris Valasek shocked the automotive industry. In their demonstration they exploited a significant vulnerability in the Uconnect infotainment system which was installed on hundreds of thousands of Fiat Chrysler Automobiles (FCA) vehicles and remotely gained control over a Jeep Cherokee while it was being driven by Wired journalist Andy Greenberg on a highway in Missouri (Greenberg, 2015).

Mr. Greenberg described this experience as terrifying; while traveling at 70 mph on an overpass, he lost the ability to accelerate and had an 18-wheeler rapidly approaching from behind him with no shoulder to steer off the road. (Henn, 2015)

The 2015 demonstration was the culmination of several years of prior research by Miller and Valasek into hacking automobiles. They had previously hacked a 2010 Ford Escape and a Toyota Prius, but only via hard wired connections; the automotive industry had dismissed those earlier demonstrations as not being significant threats because they required physical access to the automobiles in order to hack them. The 2015 demonstration changed everything; remote internet-based attacks were proved to be possible and replicable on a large scale (CISCO, 2023).

The Uconnect system was used to exploit the attack; this is a computer system with internet capabilities that controls entertainment, navigation, phone calls and Wi-Fi in vehicles (Henn, 2015). The attack had 4 phases; first was an initial access phase that was executed via the Sprint cellular network, and then different software layers of the vehicle's software were created to escalated to obtain physical access using the Controller Area Network (CAN) (Miller, 2015).

Table 1: Attack Chain Stages of the Jeep Cherokee Hack

Stage	What They Did	Targeted System	Impact Level
Stage 1	Searched the mobile network to find which Jeep Cherokees were online; like scanning for open Wi-Fi networks	Uconnect infotainment system (the car's touchscreen)	Low: They could only identify the car, not control it yet
Stage 2	Broke into the car's touchscreen system (which runs on Linux with weak security) and took control of entertainment features	Radio, air conditioning, GPS	Medium: Scary and distracting, but not yet life-threatening
Stage 3	Installed custom software onto a small	The chip connecting the screen to the	High: They now had a backdoor into the

	chip (Renesas V850) that acts as a bridge between the entertainment system and the car's physical controls	engine/brakes	car's core systems
Stage 4	Sent commands through the car's internal network (CAN bus) directly to the physical parts of the car; just like the car's own computer does	Brakes, steering, engine, transmission	Critical: Full physical control of a moving vehicle at 70 mph

Source: (Miller, 2015) (Lab, 2015)

The exploit affected more than just the Jeep Cherokee; the Uconnect platform was used on numerous vehicles produced by Fiat Chrysler between 2013 and 2015 and thus created a security flaw in all those vehicles. NHTSA provided documentation supporting the recall that indicates that there were about 1.4 million vehicles from various makes and models that are involved in the claim (NHTSA, 2015).

Table 2: Timeline of Key Events; Jeep Cherokee Hack (2013–2015)

2013	Miller & Valasek demonstrate wired hack on Ford Escape and Toyota Prius at DEF CON
Early 2015	Miller & Valasek purchase a 2014 Jeep Cherokee and begin remote vulnerability research
June 2015	Chrysler is privately notified of the Uconnect vulnerability; a software patch is prepared

July 21, 2015	Wired publishes live demonstration; public learns of the remote hack for the first time
July 23, 2015	Sprint network-level security measures implemented; FCA submits recall report to NHTSA
July 24, 2015	Fiat Chrysler announces voluntary recall of 1.4 million vehicles (Recall 15V-461)
July 2015	US Senators Markey & Blumenthal introduce the SPY Car Act in response to the incident
August 2015	Miller & Valasek present full technical details at Black Hat USA security conference

Source: (Greenberg, 2015) (NHTSA, 2015) (Lab, 2015)

3. Social Issues

The hacking of the Jeep Cherokee highlighted several important problems in our society today; including public safety, trust in technology, and society's dependency on connected systems.

3.1 Threat to Public Safety

The immediate problem addressed with the hack is the direct risk to human life. (Van der Sloot, 2022) state that as soon as a hacker gains control over a vehicle's steering and speed, the safety of all drivers using the road is at catastrophic risk; if vehicles were hacked beyond control, they could be utilized as weapons in terrorist attacks. With multiple cars hacked simultaneously, major arteries for transporting and supplying goods could be blocked, and first responders would not be able to carry out their duties effectively.

Miller and Valasek are ethical researchers who had permission from the owner of the Jeep, and therefore, acted within ethical boundaries in their demonstration but the same vulnerability left unsecured (or in the wild) would be available to any person with malicious intent and/or purpose.

3.2 Erosion of Consumer Trust

As a result of the public demonstration of the hack on the Jeep, consumer confidence in the automotive industry has suffered tremendously. HBR article (as noted by HBR) illustrates that after viewing a video of two researchers hacking rated Jeep Cherokee and demonstrating their method of doing so, questions were raised about the automotive industry and how easy it was for them to determine a breach in their security protocol! Such concerns were echoed across all automotive brands (and amplified by the media reaching a global audience via Wired magazine) about their own vehicles being compromised.

3.3 Privacy Implications

The hacking incident went beyond mere physical alteration of the car's controls; the hack brought to light major privacy concerns. According to (Lab, 2015), the researchers were able to trace the car's GPS coordinates with zero software changes to the Uconnect system; this built-in feature indicates that Uconnect enabled the car to produce sensitive location information, which any attacker with access could exploit with no security measures in place.

The violation of privacy resulting from this breach is especially concerning as it occurred during an era of increased awareness of the need to protect personal data.

3.4 National Security and Systemic Risk

The most impromptu of the social dimensions is the risk of large-scale, coordinated attacks. (Lab, 2015) cautioned that an experienced hacker could gain control of many head units and then

utilize these head units to explore the Sprint network to locate additional vulnerable vehicles, thereby establishing a type of automotive botnet consisting of hundreds of thousands of cars. The ability for nation-states and terrorist organizations to utilize similar exploits creates significant national security issues.

Table 3: Summary of Social Impacts of the 2015 Jeep Cherokee Hack

Impact Area	Description	Affected Parties
Public Safety	Remote control of brakes, engine, steering on live highway	All road users
Consumer Trust	Exposed systemic failure in connected vehicle security	FCA owners, public
Privacy	GPS tracking accessible without user consent	Vehicle owners
National Security	Potential for vehicular botnet across Sprint network	Government, citizens
Industry Reputation	Forced recall of 1.4 million vehicles globally	FCA, auto industry

Source: (Greenberg, 2015) (Lab, 2015) (Van der Sloot, 2022)

4. Ethical Issues

The Jeep Cherokee hack presents several ethical dilemmas involving responsibility, disclosure, and the balance between innovation and safety.

4.1 Responsible Disclosure

Miller and Valasek showed responsible disclosure practices when they informed Fiat Chrysler about their vulnerability before disclosing it publicly (Greenberg, 2015); (School, 2016).

Because they communicated their findings to Chrysler, within a few months, Fiat Chrysler was able to create a patch prior to widespread dissemination of the vulnerability.

Their actions demonstrate (ACM, 2018) principle 2.5 to evaluate computer systems and their effects on society, especially regarding potential risks, as well as the BCS Code of conduct when acting in the public good.

4.2 Utilitarian Analysis

The public demonstration of hacking the connected vehicle has utility. Prior to the public proof-of-concept, the automobile industry viewed earlier live demonstrations of connected car hacking as not posing a threat (CISCO, 2023).

The lack of a public demonstration would have likely prevented the voluntary recall of 1.4 million vehicles. Millions of motorists are now safe from a significant vulnerability that automotive manufacturers have ignored for years; utility is met at this point.

4.3 Deontological Analysis

On the other hand, a Kantian deontological analysis raises many ethical issues with the live demonstration on a public roadway. The hack and the associated risk were to Greenberg, and other (non-consenting) users of the roadway. While Greenberg willingly volunteered to act as a "digital crash test dummy," the other participants (motorists and pedestrians) on the roadway were not given the same opportunity to opt-in or opt-out.

According to Kant's categorical imperative, would this action be a universal rule? If all security researchers were able to conduct live highway hacking demonstrations, the risk of harming innocent third parties would be beyond question. As the experiment(s) were well-intentioned, this does not justify crossing this ethical boundary.

4.4 Manufacturer Ethical Responsibility

Fiat Chrysler has major ethical responsibility in this situation. According to Brookings Institution (Institution, 2015), a multi-layered strategy for cybersecurity must deal with both the obvious

types of exposure as well as the second-order vulnerabilities that result from the linkages between components of a system.

Therefore, connecting an entertainment system to safety-critical systems without proper isolation represents a significant failure of ethics in product design. Purchasers of connected vehicles should expect that the manufacturer has considered and mitigated anticipated cybersecurity risks.

5. Legal Issues

The Jeep Cherokee hack also highlights important legal considerations related to cybersecurity, liability, and regulatory compliance.

5.1. Existing Legal Framework at the Time

Under the National Traffic and Motor Vehicle Safety Act, the NHTSA had authority to require recalls for vehicles with safety defects. (CISCO, 2023) notes that while no mandated cybersecurity standard existed, NHTSA maintained that cybersecurity flaws threatening human life fell within its enforcement authority, since manufacturers were required to ensure systems were designed free of defects.

The hack triggered a formal NHTSA investigation into the effectiveness of FCA's recall remedy.

5.2 The Computer Fraud and Abuse Act (CFAA)

The legal status of Miller and Valasek's research was nuanced. Under the Computer Fraud and Abuse Act (CFAA), unauthorised access to computer systems is a federal offence. While the researchers had the owner's consent and had notified Chrysler in advance, the CFAA's broad language could technically encompass their activities (Greenberg, 2015).

This exposed a critical gap in US law: no clear legal safe harbour existed for good-faith security researchers conducting vulnerability testing a gap that continues to be debated in cybersecurity law reform discussions.

5.3 Regulatory and Legislative Responses

Following the incident, Senators Edward Markey and Richard Blumenthal introduced the SPY Car Act, which would have directed the NHTSA and the Federal Trade Commission to establish federal standards to secure cars and protect drivers' privacy (Henn, 2015). While this legislation was not immediately enacted, it signalled growing legislative awareness.

NHTSA subsequently issued its Cybersecurity Best Practices for Modern Vehicles in 2016; non-binding guidance encouraging risk assessments, vulnerability management, and secure over-the-air updates (NHTSA, 2015).

5.4 Supply Chain Liability

The NHTSA's investigation extended beyond FCA to Harman Kardon, the manufacturer of the Uconnect infotainment system, after determining that potentially 2.8 million systems across multiple vehicle brands could be affected.

This raised important questions about supply chain liability: to what extent does responsibility for a cybersecurity vulnerability lie with the component manufacturer rather than the vehicle assembler? Existing product liability law at the time did not provide clear answers for software-based defects in connected vehicles.

5.5 International Developments

Globally, incidents like the Jeep hack contributed to the development of UNECE WP.29 Regulation R155, which came into force in 2021 and requires vehicle manufacturers to implement cybersecurity management systems throughout the vehicle lifecycle (UNECE, 2021). (Van der Sloot, 2022) note, however, that current legislative measures focus predominantly on preventing unauthorised access, the first line of defence, while failing to address what happens after a system has been breached.

6. Professional Issues

From a professional perspective, the Jeep Cherokee hack underscores the importance of adhering to industry standards, ethical guidelines, and best practices in cybersecurity.

6.1 The Researchers' Professional Conduct

Charlie Miller and Chris Valasek demonstrated significant professional integrity in their approach. By notifying Chrysler in advance and publishing their findings only after a patch was available, they adhered to the principle of responsible disclosure aligned with ACM Code of Ethics principle 2.5 (give comprehensive evaluations with special emphasis on risks) and BCS Code of Conduct section 2 (act in the public interest).

Their decision to present at Black Hat USA 2015 ensured the security community could learn from the research without providing a 'how-to' guide for malicious actors.

6.2 Fiat Chrysler's Engineering Failures

The core professional failure in this incident was Fiat Chrysler's approach to secure software design. IEEE Code of Ethics Principle 1 requires engineers to prioritise public safety, health, and welfare (IEE, 2020). (International, 2021), the automotive cybersecurity engineering standard, requires risk-based identification and protection of safety-critical systems throughout the development lifecycle.

The fact that an entertainment system could access the CAN bus controlling brakes and steering represented a fundamental violation of secure-by-design principles. The architecture placed commercial feature development above safety engineering.

6.3 Industry-Wide Professional Responsibility

The Jeep hack exposed a systemic failure across the automotive industry. As (CISCO, 2023) observed, the NHTSA's cybersecurity best practice guidelines emphasise that engaged leadership is the single most important element of a successful cybersecurity programme.

In the rush to compete on connectivity features, manufacturers across the industry had largely failed to employ dedicated cybersecurity teams or conduct formal threat modelling before deploying internet-connected systems in vehicles.

7. Conclusion and Personal Reflection

7.1 Conclusion

The 2015 Jeep Cherokee hack proved that connected cars can be the victims of sophisticated cyber-attacks that may pose a safety threat to humans. This incident exposed significant gaps in automotive cybersecurity, such as system isolation issues, lack of risk assessment, and security testing.

The hack had social, ethical, legal and professional implications. Socially, it raised concerns about privacy, public safety and national security. Ethically, it posed questions of responsible disclosure, consent and manufacturer responsibilities. It brought to light legal shortcomings in the cyber security regulation and liability frameworks. Technically, it highlighted the importance of achieving cybersecurity in product development by the engineers and organizations.

In conclusion, the incident had a significant impact on the automotive industry, inspiring greater focus on cybersecurity standards, regulations, and the risks of connected systems.

7.2 Personal Reflection

This Jeep Cherokee hack offers a valuable insight into the interplay between technology, ethics and public safety. While modern digital systems offer significant advantages to society, they also present novel risks which need to be managed with care.

Brainstorming and analysis phases should be performed before organisations deploy connected technologies from the standpoint of ethical decision making. In the brainstorming process, developers should consider all potential risks, stakeholders, and consequences that could occur from system vulnerabilities. Organisations should consider ethical theories, legal obligations and public safety implications before deciding on deployment during the analysis phase.

The incident shows that cybersecurity is not something you can afford to overlook. Instead, it should be part of the overall design, development, testing, and maintenance of the system.

Car companies should follow the following steps: implement secure-by-design principles, regular software updates, network segmentation, and continuous penetration testing and more robust encryption. Governments need to keep advancing cybersecurity policies and universities and professional bodies need to further develop cybersecurity education and ethics training.

As the Jeep Cherokee incident proves, without adequate cybersecurity, technology can be a major threat to society. Future connected technologies must thus continue to be developed in the context of ethical and professional responsibility.

8. References

ACM, 2018. [Online]

Available at: <https://www.acm.org/code-of-ethics>

CISCO, F., 2023. [Online]

Available at: <https://fractionalciso.com/the-groundbreaking-2015-jEEP-hack-changed-automotive-cybersecurity/>

Greenberg, A., 2015. *Hackers Remotely Kill a Jeep on the Highway*. [Online]

Available at: <https://www.wired.com/2015/07/hackers-remotely-kill-jeep-highway/>

Henn, S., 2015. *The Ghost in the Car May Be a Hacker*. [Online]

Available at: <https://www.npr.org/sections/alltechconsidered/2015/07/21/424988397/the-ghost-in-the-car-may-be-a-hacker>

IEE, 2020. [Online]

Available at: <https://www.ieee.org/about/corporate/governance>

Institution, B., 2015. [Online]

Available at: <https://www.brookings.edu/articles/jeep-cherokee-hack-offers-important-lessons-on-the-security-of-things/>

International, I. & S., 2021. [Online]

Available at: <https://www.iso.org/standard/70918.html>

Lab, K., 2015. [Online]

Available at: <https://www.kaspersky.com/blog/blackhat-jeep-cherokee-hack-explained/9493/>

Miller, C. & V. C., 2015. [Online]

Available at: <http://illmatics.com/Remote%20Car%20Hacking.pdf>

NHTSA, 2015. [Online]

Available at: <https://static.nhtsa.gov/odi/inv/2015/INOA-RQ15004-8729.PDF>

School, H. B., 2016. [Online]

Available at: <https://d3.harvard.edu/platform-rctom/submission/dude-where-s-my-car-car-hacking-in-the-digital-age/>

UNECE, 2021. [Online]

Available at: <https://unece.org/transport/documents/2021/03/standards/un-regulation-no-155-cyber-security-and-cyber-security>

Van der Sloot, B. & W. A., 2022. [Online]

Available at: <https://www.tandfonline.com/doi/full/10.1080/13600869.2022.2060472>